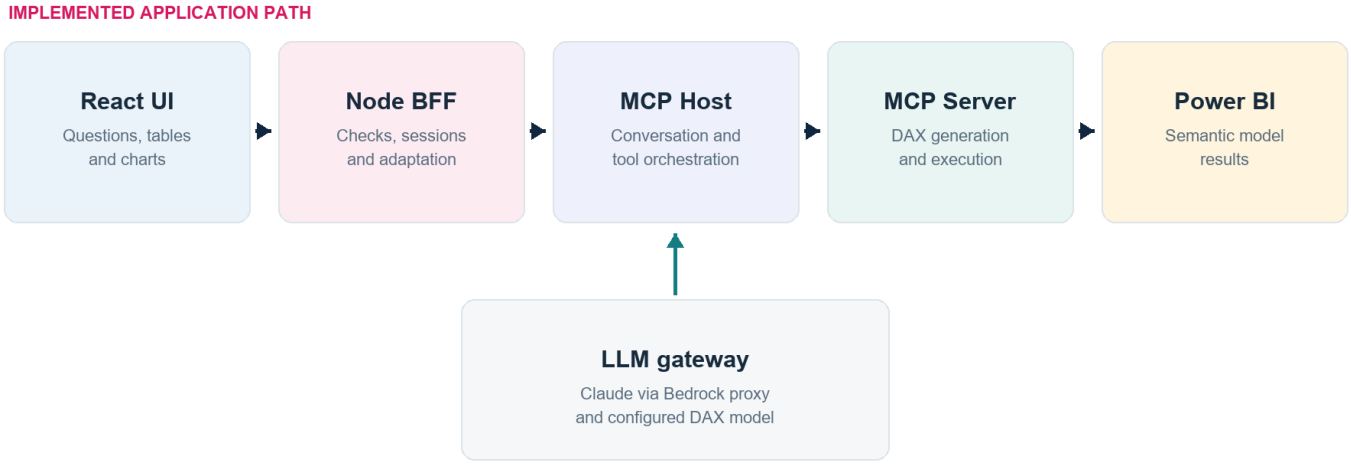
Nielsen Conversational BI

A conversational route from questions to Power BI evidence

The solution combines a React chat experience, a Node.js backend for frontend (BFF), and an existing Python MCP analytics pipeline. It turns a question about a selected semantic model into an analytical answer with structured data and visualizations where supported.

Solution architecture



The BFF adapts the browser request to the MCP Host's HTTP chat contract. The Host coordinates the conversation and tool call; the MCP Server generates DAX and queries Power BI. The result returns through the same layers for presentation.

What the application layer adds

One workspace

Model picker, chat, tables, charts and CSV export.

Controlled handoff

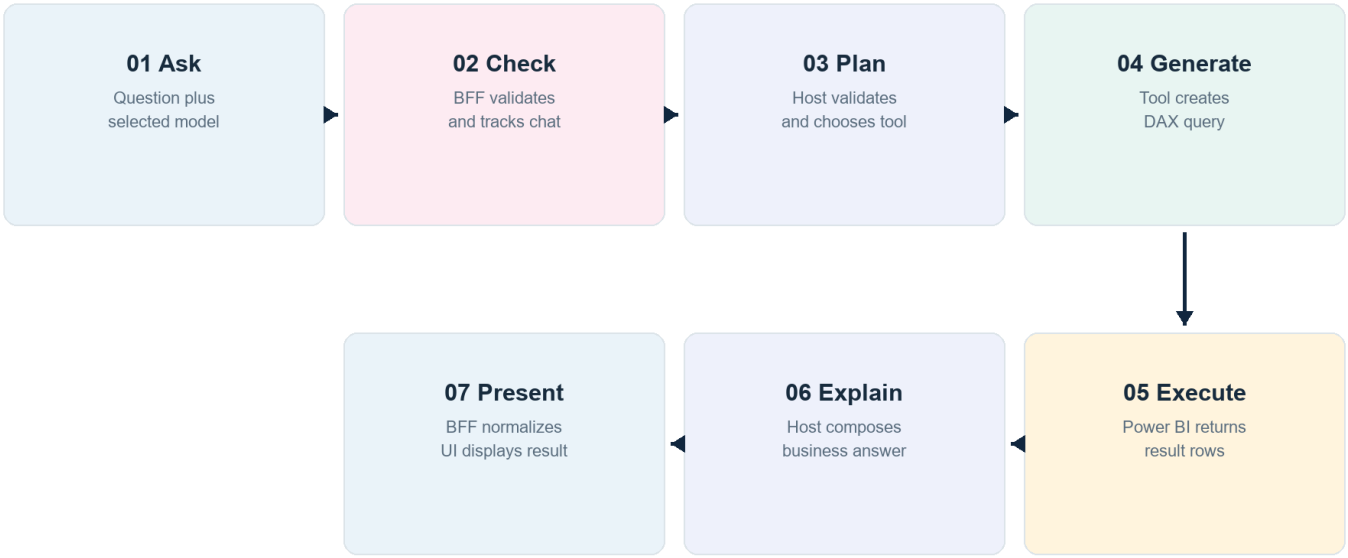
Request checks, MCP mapping and response normalization.

Reusable layers

UI, BFF and analytics services evolve separately.

From question to answer

The current interface asks the user to select a semantic model. The services then validate, generate, execute and present the result without exposing the analytics tool directly to the browser.



Narrative, result tables and visualizations return through the same UI.

Reporting experience

A successful response can include a plain-language finding, result rows and a chart. The UI renders tables and supported visualizations; tabular results can be exported as CSV. This is an analytical answer workflow, not a scheduled PDF reporting service.

How model choice works today

The UI contains two bundled model definitions. The BFF supports a configured catalogue and model ID mapping, but an empty catalogue does not enforce an allowlist. Adding a semantic model therefore needs a deliberate configuration and integration check; automatic discovery is a separate enhancement.

Security and service boundaries

The reviewed no-Entra application validates request shape and content at the BFF, limits and normalizes MCP output, and keeps outbound credentials on the server. It does not authenticate individual UI users: every BFF request receives one shared workspace identity.

Current boundary and production extension

Access	Current: entry requires an approved company gateway or private network; there is no inbound Entra sign-in.	Target: verified user identity and per-model authorization.
Data	Current: BFF checks prompts and normalizes output; the Power BI gateway uses a configured service principal.	Target: documented least privilege and query audit controls.
History	Current: UI history is browser-local; BFF and Host session state is process-local, with a shared BFF owner.	Target: durable history tied to verified user identity.

Why the layers matter

The UI never calls the analytics tool directly. The BFF holds the browser-facing contract, the MCP Host manages conversation and tool use, and the MCP Server owns DAX execution. Each boundary can have its own access policy, logging, timeout and release process.

Decision for external deployment

Before opening the application beyond a controlled pilot, add trusted sign-in and per-user session ownership. A browser-generated UUID helps correlate requests but is not proof of identity or permission to retrieve another conversation.

Upgrade and resilience path

The frontend and BFF build as separate images, with health endpoints and configurable MCP field mapping. High availability and recoverable history, however, require shared infrastructure beyond the current process-local stores.

Recommended production sequence

01 **Secure access**

Add verified identity and enforce model and session ownership.

02 **Persist state**

Move chat history, Host context and rate limits out of process memory.

03 **Scale services**

Run multiple healthy replicas and use controlled, reversible rollouts.

04 **Prove recovery**

Back up durable state and test a full restore with configuration.

Availability and backup design

For higher availability, place stateless UI and BFF replicas behind an ingress and use shared storage for sessions and rate limits. Define recovery objectives, back up the durable history store and configuration, then rehearse restoration. Container restarts and image copies do not back up in-memory chats.

Technology references

Model Context Protocol architecture

Amazon Bedrock Claude guidance

Power BI semantic models

Kubernetes deployments

Kubernetes autoscaling

OWASP session management